

Introduction to Machine Learning

Homework 3

Songchen Wang Student ID: 2024201594

May 8, 2026

Question 1

Robust NMF under Laplace noise. Assuming i.i.d. Laplace noise on entries, the negative log-likelihood is proportional to an ℓ_1 data-fitting term. Given $X \in \mathbb{R}_+^{N \times D}$, robust NMF solves

$$\min_{U \geq 0, V \geq 0} \|X - UV^\top\|_1. \quad (1)$$

IRLS idea. We solve (1) by iteratively reweighted least squares (IRLS). Given residual $R^{(t)} \triangleq X - U^{(t)}(V^{(t)})^\top$, define element-wise weights (with $\varepsilon > 0$)

$$W_{ij}^{(t)} \triangleq \frac{1}{|R_{ij}^{(t)}| + \varepsilon}. \quad (2)$$

Then we approximately solve

$$\min_{U \geq 0, V \geq 0} \sum_{i,j} W_{ij}^{(t)} (X_{ij} - (UV^\top)_{ij})^2. \quad (3)$$

Multiplicative updates (weighted NMF). Fix W and define the weighted objective

$$f(U, V) \triangleq \sum_{i,j} W_{ij} (X_{ij} - (UV^\top)_{ij})^2, \quad (4)$$

which is the same as (3). Using matrix calculus, the gradients are

$$\nabla_U f = -2(W \odot X)V + 2(W \odot (UV^\top))V, \quad (5)$$

$$\nabla_V f = -2(W \odot X)^\top U + 2(W \odot (UV^\top))^\top U. \quad (6)$$

Under the nonnegativity constraints, a KKT-style element-wise condition is

$$U \geq 0, \nabla_U f \geq 0, U \odot \nabla_U f = 0 \quad (\text{and similarly for } V). \quad (7)$$

Split the gradient into a “negative part” and “positive part”, $\nabla_U f = 2(\nabla_U^+ - \nabla_U^-)$ with $\nabla_U^- = (W \odot X)V$ and $\nabla_U^+ = (W \odot (UV^\top))V$ (analogously for V). This yields the multiplicative updates

$$U \leftarrow U \odot \frac{(W \odot X)V}{(W \odot (UV^\top))V + \delta}, \quad (8)$$

$$V \leftarrow V \odot \frac{(W \odot X)^\top U}{(W \odot (UV^\top))^\top U + \delta}, \quad (9)$$

where divisions are element-wise and $\delta > 0$ avoids division by zero.

Algorithm 1 Robust NMF via IRLS (Laplace noise)

Require: $X \in \mathbb{R}_+^{N \times D}$, rank L , iterations T , $\varepsilon, \delta > 0$

- 1: Initialize $U \in \mathbb{R}_+^{N \times L}$, $V \in \mathbb{R}_+^{D \times L}$ (e.g., random nonnegative)
- 2: **for** $t = 0, 1, \dots, T - 1$ **do**
- 3: $R \leftarrow X - UV^\top$
- 4: $W \leftarrow 1/(|R| + \varepsilon)$ (element-wise)
- 5: Update U by (8); update V by (9)
- 6: **end for**

Ensure: Nonnegative factors U, V ; reconstruction $\hat{X} = UV^\top$

Question 2

Many spectral/manifold methods can be written in the unified form

$$\min_{Z \in \mathbb{R}^{N \times d}} \text{tr}(Z^\top \Phi Z), \quad \text{s.t. } Z \in \Omega, \quad (10)$$

where rows of Z are the low-dimensional coordinates, $\Phi \in \mathbb{R}^{N \times N}$ is symmetric, and Ω removes trivial solutions.

(1) Laplacian Eigenmaps. Let $S \in \mathbb{R}^{N \times N}$ be an affinity matrix, $D \triangleq \text{diag}(S\mathbf{1})$, and $L \triangleq D - S$. Eigenmaps minimizes the smoothness energy

$$\sum_{i,j} S_{ij} \|z_i - z_j\|_2^2 = 2 \text{tr}(Z^\top LZ), \quad (11)$$

so it matches (10) with $\Phi = L$. To avoid the trivial minimizer, a standard feasible set is

$$\Omega_{\text{LE}} \triangleq \{Z : Z^\top DZ = I, Z^\top D\mathbf{1} = 0\}. \quad (12)$$

(2) LLE. Let $W \in \mathbb{R}^{N \times N}$ be the reconstruction weights (each row sums to 1). The embedding is obtained by

$$\min_Z \sum_i \left\| z_i - \sum_j W_{ij} z_j \right\|_2^2 = \|(I - W)Z\|_F^2 = \text{tr}(Z^\top MZ), \quad (13)$$

where $M \triangleq (I - W)^\top(I - W)$, hence $\Phi = M$. A common feasible set is

$$\Omega_{\text{LLE}} \triangleq \{Z : Z^\top Z = I, Z^\top \mathbf{1} = 0\}. \quad (14)$$

(3) Kernel PCA.

Let K be the Gram matrix and $H \triangleq I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$.

After centering, $K_c \triangleq HKH$.

Kernel PCA maximizes projected variance $\text{tr}(Z^\top K_c Z)$ with $Z^\top Z = I$, which is equivalent to

$$\min_Z \text{tr}(Z^\top (-K_c)Z), \quad \Omega_{\text{kPCA}} \triangleq \{Z : Z^\top Z = I\}, \quad (15)$$

so $\Phi = -K_c$.

(4) **ISOMAP.** Let D^{geo} be the matrix of graph-based geodesic distances. Let $C \triangleq I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ be the centering matrix and let D^{geo} be the matrix of graph-based geodesic distances. Classical MDS constructs the inner-product (kernel) matrix

$$K \triangleq -\frac{1}{2}C(D^{geo} \circ D^{geo})C. \quad (16)$$

ISOMAP maximizes $\text{tr}(Z^\top K Z)$ subject to $Z^\top Z = I$ (and typically $Z^\top \mathbf{1} = 0$), equivalently

$$\min_Z \text{tr}(Z^\top (-K) Z), \quad \Omega_{\text{ISO}} \triangleq \{Z : Z^\top Z = I, Z^\top \mathbf{1} = 0\}, \quad (17)$$

hence $\Phi = -K$.

Method	Φ	Ω
Eigenmaps	$L = D - S, D = \text{diag}(S\mathbf{1})$	$Z^\top D Z = I, Z^\top D \mathbf{1} = 0$
LLE	$M = (I - W)^\top (I - W)$	$Z^\top Z = I, Z^\top \mathbf{1} = 0$
Kernel PCA	$-K_c, K_c = H K H, H = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$	$Z^\top Z = I$
ISOMAP	$-K, K = -\frac{1}{2}C(D^{geo} \circ D^{geo})C$	$Z^\top Z = I, Z^\top \mathbf{1} = 0$

Table 1: Summary of Φ and Ω for (10).

All four methods reduce to (10); the embedding is obtained from eigenvectors of Φ under the stated constraints (equivalently, top eigenvectors of $-\Phi$ for maximization forms).